CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · FORMULA BANK

# Chapter 1: Rational and Irrational Numbers

Every formula and result of the chapter on one sheet

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|---|---|---|
| 1 | Rational number | $\frac{p}{q}$, $p$ and $q$ integers, $q \neq 0$ |
| 2 | Terminating decimal | Denominator (lowest terms) has only the prime factors 2 and 5 |
| 3 | Rational between $a$ and $b$ | $\frac{a+b}{2}$ |
| 4 | Irrational number | Decimal neither terminates nor recurs |
| 5 | Product rule | $\sqrt{ab} = \sqrt{a}\sqrt{b}$ |
| 6 | Quotient rule | $\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$ |
| 7 | Conjugate product | $(a+\sqrt{b})(a-\sqrt{b}) = a^2 - b$ |
| 8 | Conjugate product | $(\sqrt{a}+\sqrt{b})(\sqrt{a}-\sqrt{b}) = a - b$ |
| 9 | Square of a sum | $(\sqrt{a}+\sqrt{b})^2 = a + b + 2\sqrt{ab}$ |
| 10 | Rationalising $\frac{1}{\sqrt{a}}$ | $\frac{\sqrt{a}}{a}$ |
| 11 | Rationalising $\frac{1}{\sqrt{a}+\sqrt{b}}$ | $\frac{\sqrt{a}-\sqrt{b}}{a-b}$ |
| 12 | Useful values | $\sqrt{2} \approx 1.414$, $\sqrt{3} \approx 1.732$, $\sqrt{5} \approx 2.236$ |